

A Planar Counterexample to Absolute Leray-Kernel Boundedness on Strongly $\mathbb{C}$ -Convex Domains

Candidate counterexample packet for arXiv:2506.02913

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Status. Candidate full negative answer to Remark 7.3 as literally stated; likely valid, pending human review. The question quantifies over $D \subset \mathbb{C}^n$ without imposing $n \geq 2$. In dimension one, strong $\mathbb{C}$ -convexity is vacuous beyond ordinary $\mathbb{C}$ -convexity, and an explicit smooth radial domain has an interior zero of the denominator B_D . The positive kernel is then nonintegrable on an open set. If the intended question silently assumes $n \geq 2$, this packet does not settle that strengthened version.

1 Source question

Chatterjee, *Dual realizations of Bergman spaces on strongly convex domains*, arXiv:2506.02913v2 [1], proves L^p boundedness of the positive operator $|\mathcal{B}_D|$ when D is strongly real convex. Remark 7.3 asks:

Let $D \subset \mathbb{C}^n$ be a bounded $\mathcal{C}^2$ -smooth strongly $\mathbb{C}$ -convex domain containing the origin, and let $|\mathcal{B}_D|$ be the operator in Lemma 3.1. Is $|\mathcal{B}_D|$ an $L^p(D)$ -bounded operator?

convex domain, and due to $\mathbb{C}$ -convexity, it is L^p -bounded, which is strongly convex. This shows that the hypothesis of Theorem 1.1 is satisfied for the dual complement of bounded $\mathcal{C}^2$ -smooth strongly convex domains containing the origin. For example, Ω_n^* , where Ω_n is as defined in (7.1), satisfies the hypothesis of Theorem 1.1.

Remark 7.3. The hypothesis in Theorem 1.1 of Ω^* being strongly convex is used only to show that $\mathcal{F}_{n+1}$ is a L^p -bounded operator between $A^p(\Omega)$ and $A^p(\Omega^*)$, which essentially follows from the fact that when D is a bounded $\mathcal{C}^2$ -smooth strongly convex domain containing the origin, the operator $|\mathcal{B}_D|$, as given in Lemma 3.1 is bounded on $L^p(D)$. Consequently, the requirement of Ω^* being strongly convex can be relaxed if one can answer the following question.

Question. Let $D \subset \mathbb{C}^n$ be a bounded $\mathcal{C}^2$ -smooth strongly $\mathbb{C}$ -convex domain containing the origin, and $|\mathcal{B}_D|$ is the operator as given in Lemma 3.1. Is $|\mathcal{B}_D|$ a $L^p(D)$ -bounded operator?

Remark 7.4. Note that the domain considered in Theorem 1.3 is a bounded convex Reinhardt domain which lacks $\mathcal{C}^1$ -smoothness, which raises the following question.

Question. Does there exist a bounded $\mathcal{C}^1$ -smooth convex domain $\Omega \subset \mathbb{C}^n$ such that $\mathcal{F}_{n+1}$ is not

Figure 1: Remark 7.3 and the exact question on source PDF page 27.

For a star-shaped domain with Minkowski functional m_D , the source defines

$$B_D(\zeta, z) = m_D(\zeta) 2\partial m_D(\zeta)(\zeta - z) + 1 - m_D(\zeta)^2, \quad (1)$$

$$|\mathcal{B}_D|f(z) = \int_D f(\zeta) |B_D(\zeta, z)|^{-n-1} dV(\zeta). \quad (2)$$

The pairing in (1) is ordinary multiplication when $n = 1$.

2 Proof intuition

The strongly real-convex proof uses the sign of the real supporting hyperplane to keep B_D away from zero in $D \times D$. Complex convexity controls only the modulus of a boundary complex support function; it does not give that sign. In one complex dimension there are no nonzero complex tangent directions at all. We therefore choose a smooth two-lobed radial domain. At a carefully selected point on one inward-facing side, the normal ray re-enters the opposite lobe. Moving partway down the original radial segment and then along that normal gives a second interior point at which B_D vanishes exactly.

The zero is transverse. Hence the implicit-function theorem supplies an interior zero in the ζ variable for every z in an open neighborhood. Since $n + 1 = 2$, the local kernel dominates $|\zeta - \zeta(z)|^{-2}$, whose area integral diverges logarithmically.

3 The domain and its gauge

For $\theta \in \mathbb{R}$ set

$$R(\theta) = 1 + \frac{13}{20} \cos(2\theta), \quad D = \{re^{i\theta} : 0 \leq r < R(\theta)\}. \quad (3)$$

Because $R \geq 7/20$, this is a bounded C^∞ radial domain containing 0, star-shaped with respect to 0, with only non-radial tangents. It is simply connected, hence $\mathbb{C}$ -convex in dimension one. Moreover, for any $w \in \partial D$, the complex tangent space

$$H_w \partial D = \{v \in \mathbb{C} : \partial \rho(w)v = 0\}$$

is $\{0\}$. Thus the usual condition that the second fundamental form be positive definite on $H_w \partial D$ holds vacuously. Accordingly, D is strongly $\mathbb{C}$ -convex under the standard dimension-one convention used by the question as written.

Its Minkowski functional is

$$m_D(re^{i\theta}) = \frac{r}{R(\theta)}. \quad (4)$$

If $\eta = R(\theta)e^{i\theta} \in \partial D$, polar differentiation gives

$$q(\theta) := 2\partial m_D(\eta) = e^{-i\theta} \left(\frac{1}{R(\theta)} + i \frac{R'(\theta)}{R(\theta)^2} \right). \quad (5)$$

Positive homogeneity also gives $2\partial m_D(s\eta) = q(\theta)$ for every $s > 0$.

4 An exact transverse interior zero

Theorem 1. *For the domain (3), $|\mathcal{B}_D|$ fails to map $L^p(D)$ into finite-valued functions for every $1 \leq p \leq \infty$. In particular it is not bounded on $L^p(D)$ for any $1 < p < \infty$.*

Proof. Choose θ_0 in the second quadrant by

$$\cos \theta_0 = -\frac{1}{5}, \quad \sin \theta_0 = \frac{2\sqrt{6}}{5}. \quad (6)$$

Then

$$\cos(2\theta_0) = -\frac{23}{25}, \quad \sin(2\theta_0) = -\frac{4\sqrt{6}}{25}, \quad R(\theta_0) = \frac{201}{500}.$$

Put $s = 1/2$, $\eta_0 = R(\theta_0)e^{i\theta_0}$, $\zeta_0 = s\eta_0$, and abbreviate $q_0 = q(\theta_0)$. Define

$$z_0 = \zeta_0 + \frac{1 - s^2}{sq_0}. \quad (7)$$

Both points are nonzero, and $m_D(\zeta_0) = s < 1$. Formulae (3)–(7) simplify to

$$z_0 = \frac{8811237}{43873750} + i \frac{4319088\sqrt{6}}{21936875}. \quad (8)$$

If $\phi_0 = \arg z_0$, exact arithmetic gives

$$|z_0|^2 = \frac{598702419}{2193687500}, \quad R(\phi_0) = \frac{140986860969}{260066040500}, \quad (9)$$

and hence

$$m_D(z_0)^2 = \frac{|z_0|^2}{R(\phi_0)^2} = \frac{456889764289102564}{492000073411354561} < 1. \quad (10)$$

Thus $z_0 \in D$. By construction and (1),

$$B_D(\zeta_0, z_0) = sq_0(\zeta_0 - z_0) + 1 - s^2 = 0.$$

It remains to check that this is not a harmless isolated coincidence. Write $\zeta = te^{i\theta}$ near ζ_0 and regard $B_D(\zeta, z_0)$ as a map from the real variables (t, θ) to $\mathbb{C} \simeq \mathbb{R}^2$. Direct differentiation of (1), (4), and (5) at $(t, \theta) = (sR(\theta_0), \theta_0)$ gives

$$\begin{aligned} \partial_t B_D &= -\frac{1000}{201} + i \frac{26000\sqrt{6}}{40401}, \\ \partial_\theta B_D &= \frac{1825018\sqrt{6}}{7054899} - i \frac{8776}{35099}. \end{aligned}$$

The corresponding real Jacobian is

$$\det \frac{\partial(\operatorname{Re} B_D, \operatorname{Im} B_D)}{\partial(t, \theta)} = \frac{23285456000}{95008324833} > 0. \quad (11)$$

Since $t = sR(\theta_0) > 0$, the change from polar to Cartesian coordinates is nonsingular, so $D_\zeta B_D(\zeta_0, z_0)$ is invertible as a real map.

The real implicit-function theorem now yields neighborhoods $z_0 \in V \Subset D$ and $\zeta_0 \in U \Subset D$ and a C^1 map $z \mapsto \zeta(z) \in U$ such that

$$B_D(\zeta(z), z) = 0 \quad (z \in V).$$

After shrinking the neighborhoods, the first derivatives of B_D are uniformly bounded there. Therefore, for some $C > 0$,

$$|B_D(\zeta, z)| \leq C|\zeta - \zeta(z)|$$

whenever $z \in V$ and ζ lies in a fixed small disk about $\zeta(z)$ contained in U . Taking $f \equiv 1 \in L^p(D)$ and recalling that $n = 1$,

$$|\mathcal{B}_D|1(z) \geq C^{-2} \int_{|\zeta - \zeta(z)| < r} |\zeta - \zeta(z)|^{-2} dA(\zeta) = \infty, \quad z \in V.$$

Thus the output is infinite on an open set, proving the theorem. $\square$

5 Why the real-convex argument breaks

The source first uses real convexity to show that the real tangent line at a boundary point does not meet $\overline{D}$ elsewhere. That makes the real part of B_D vanish only on the boundary diagonal. It then upgrades this to a two-sided nonisotropic estimate and applies a weighted Schur test. For (3), the outward normal from one side enters the opposite lobe; the point (7) converts precisely that geometry into the zero $B_D(\zeta_0, z_0) = 0$. No Schur estimate can survive an interior transverse zero of the positive kernel.

6 Verification, scope, and novelty

The accompanying exact-arithmetic script independently checks (8)–(11) and the identity $B_D(\zeta_0, z_0) = 0$. No floating-point inequality is used in the proof. The source PDF is copied locally and the question crop is taken from page 27.

The scope qualification is essential. This is a full counterexample to the quantified question printed in Remark 7.3 because that question does not exclude $n = 1$. Several-complex-variables authors sometimes reserve the phrase “strongly $\mathbb{C}$ -convex” for $n \geq 2$, precisely because the complex tangent condition is empty in dimension one. Under that intended reinterpretation, the packet proves only that an $n \geq 2$ hypothesis is necessary; it does not answer the higher-dimensional problem.

A bounded novelty search on 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact-title and exact-question web searches; arXiv id 2506.02913; and the phrases “strongly $\mathbb{C}$ -convex”, “ B_D ”, “Leray kernel”, and “ L^p bounded”. It found the source, general Cauchy–Leray literature, and summaries of the source, but no later paper answering Remark 7.3 and no occurrence of this radial counterexample. This supports novelty only within those stated bounds.

Human-review recommendation. Verify first that the intended convention in Remark 7.3 permits $n = 1$. Conditional on that literal reading, the proof reduces to exact arithmetic and a standard implicit-function argument and should be checked as a candidate full negative answer. If the authors intended $n \geq 2$, classify it instead as a sharp dimensional caveat and retain the higher-dimensional question as open.

References

- [1] A. Chatterjee, *Dual realizations of Bergman spaces on strongly convex domains*, arXiv:2506.02913v2, 2026, Remark 7.3 and Lemma 3.1.